H.T.No: | | | | | | | | | | | Course Code: **241EE001**

ADITYA UNIVERSITY

B.Tech – II Semester End Examinations Regular – Jun 2025

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
(Common to IT, PT, AIML & CSE(DS))

Time: 3 hours **Max. Marks: 100**

Answer ONE question from each unit
All Questions Carry Equal Marks
All parts of the questions must be answered at one place only

UNIT-I

1	a	Explain Kirchhoff's Current Law and illustrate its application in solving a circuit.	L2	CO1	[8M]
	b	In the resistance circuit of Fig.1, find v_1 and i_2.	L3	CO1	[12M]

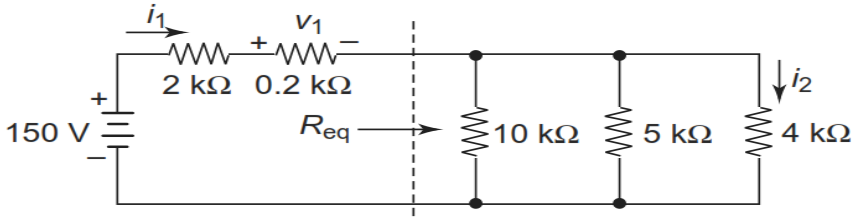


Fig.:1

(OR)

2	a	Explain the following terms i)Cycle ii) Amplitude iii) Time period iv) Frequency v) Instantaneous value.	L2	CO1	[10M]
	b	Find the values of different voltages that can be obtained from 25V source with the help of voltage divider circuit of Fig.2	L3	CO1	[10M]

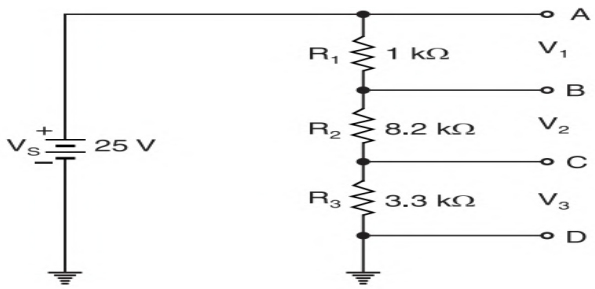


Fig.:2

UNIT-II

3	a	Explain the operation of DC motor with neat diagram.	L2	CO2	[10M]
	b	Explain the working of attraction type moving iron instrument	L2	CO2	[10M]

(OR)

4	a	Explain the operation of Alternator with neat diagram.	L2	CO2	[10M]
	b	Explain the working of wheatstone bridge with neat diagram	L2	CO2	[10M]

(P.T.O)

UNIT-III

5	a	Explain the Layout and operation of Thermal Power Plant.	L2	CO3	[10M]
	b	What are the causes of electric shock, and how can they be avoided?	L1	CO3	[10M]
		(OR)			
6	a	What are the essential components of a wind turbine? Explain their functions.	L2	CO3	[10M]
	b	Explain the working principle of Miniature circuit breaker with neat diagram.	L2	CO3	[10M]

UNIT-IV

7	a	Explain CE configuration of BJT.	L2	CO4	[10M]
	b	Draw and explain the V-I characteristics of a PN junction diode in forward and reverse bias.	L2	CO4	[10M]
		(OR)			
8	a	Explain the operation of Zener diode with neat sketch.	L2	CO4	[10M]
	b	Describe the operation of NPN transistor.	L3	CO4	[10M]

UNIT-V

9	a	Explain the working of a half-wave rectifier with the help of a circuit diagram and waveforms.	L2	CO5	[10M]
	b	Explain the C filter with neat diagram	L2	CO5	[10M]
		(OR)			
10		Write the truth tables for NOT, OR, AND, NOR, NAND, XOR, and XNOR gates.	L2	CO5	[20M]

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